

15) Limitations, Risks and Future Roadmap

Last updated by | Mennell, Jacob | 19 May 2026 at 15:40 BST

Limitations, Risks, And Roadmap

Purpose

This page captures the current technical boundaries visible in the solution, the main operational and AI-related risks, and the most natural future improvement areas.

Scope

The goal of this page is not to criticize the solution, but to make its current tradeoffs explicit. For an externally shared technical wiki, credibility depends on acknowledging boundaries as well as strengths.

Current Limitations

Retrieval quality depends on ingestion quality

The retrieval stack is only as good as the chunks produced during ingestion. Weak document structure, sparse metadata, or fallback ingestion paths can reduce the usefulness of both semantic and hybrid retrieval.

Lexical and semantic tradeoffs remain real

Lexical retrieval performs well for literal terminology and abbreviations, while semantic retrieval helps when wording varies. Neither is universally dominant, which is why the solution uses hybrid retrieval and reranking rather than depending on a single signal.

Progressive orchestration adds complexity

Progressive-stop and progressive-fanin behaviors are powerful, but they are also more operationally complex than a simple single-pass extraction model. They require stronger observability and more careful QA tuning.

AI outputs remain probabilistic

Even with retrieval, routing, and QA, extraction and checks are not purely deterministic business rules. The architecture reduces risk through bounded retries and fail-soft outputs, but it does not remove model uncertainty entirely.

Operational Risks

- ingestion failure or degraded parsing quality can reduce the evidence available to later workflows
- queue or worker issues can delay asynchronous orchestration
- over-broad retrieval can introduce noise into downstream reasoning
- under-broad retrieval can omit the evidence needed for a correct answer
- retry policies that are too aggressive can increase cost and complexity without improving outcomes

Architecture Risks To Monitor

- rising QA retry rates
- increasing percentages of fail-soft outputs such as `stuck`

- low retrieval confidence or poor evidence attribution
- operational dependence on complex multi-stage orchestration for simple questions

These are not necessarily current failures, but they are the right signals to watch as the system evolves.

Natural Roadmap Directions

Based on the repository's current structure, the most natural improvement areas are:

1. stronger retrieval observability

More explicit debugging and telemetry around why specific chunks won, lost, or were reranked.

2. richer ingestion enrichment

Additional improvement to how chunk text and structure capture headers, tables, and surrounding context.

3. tighter task-to-retrieval alignment

Continued improvement to how retrieval query shape, rerank query shape, and execution mode interact for each task type.

4. clearer operational dashboards

Better summarized views of QA outcomes, retry behavior, and workflow health.

5. documentation and explainability maturity

Continued improvement to externally shareable source attribution, architectural diagrams, and operational guidance.

Why These Limits Are Acceptable In Context

The solution already includes several strong design responses to these risks:

- strict retrieval scoping
- multiple retrieval strategies
- bounded retry behavior
- fail-soft handling
- dedicated tests around retrieval signals
- explicit architecture notes for complex workflow behavior

In other words, the solution does not ignore its harder problems; it addresses them with structured engineering controls.